

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Health and Safety Systems in the Oil and Gas Post Covid

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Health and Safety Management Systems (HSMS) are essential for ensuring safe operations in high-risk industries such as oil and gas

- Traditionally, HSMS relied on:

- Physical inspections and on-site supervision
- Direct communication and face-to-face safety monitoring
- Stable workforce structures and centralized decision-making

- The COVID-19 pandemic acted as an unprecedented disruption, forcing:

- Remote monitoring and limited site access
 - Reduced workforce presence and operational adjustments
 - This crisis exposed major limitations in traditional safety systems and required rapid transformation
- Safety definition expanded beyond physical hazards to include:
- Mental health
 - Digital adaptability
 - Workforce resilience

Objectives

- To evaluate the **resilience and adaptability** of HSMS across lockdown, transition, and recovery phases
- To identify structural weaknesses in traditional safety systems under crisis conditions
- To analyze behavioral changes such as:
 - Leadership transformation
 - Increased individual accountability
- To assess the role of **digital tools (IoT, remote monitoring)** in maintaining safety continuity
- To examine the impact of workforce fragmentation on safety performance
- To explore the relationship between **mental well-being and operational safety**
- To study decision-making challenges under uncertainty
- To identify best practices adopted during the pandemic
- To develop a **future-ready, resilient HSMS framework**

- Adopted a **qualitative case study approach** to capture real-world safety system behavior
- Focused on understanding human experiences and system adaptations during crisis
- Used **triangulation technique** for reliable and validated data collection
- Data Collection Methods:
 - **Narrative Interviews**
 - Conducted with safety managers, engineers, and frontline workers
 - Provided detailed insights into operational challenges
 - **Online Surveys**
 - Collected data from a larger workforce
 - Measured stress levels, digital adoption, and safety awareness
 - **Document Analysis**
 - Reviewed SOPs, maintenance logs, and incident reports
 - Compared planned vs actual practices
 - Analysis Framework

- Data Collection Tools:
 - Virtual communication platforms for conducting interviews
 - Online survey tools for large-scale data collection
 - Document management systems for accessing reports and SOPs
- Data Analysis Tools:
 - Qualitative coding techniques for organizing data
 - Thematic analysis frameworks for identifying patterns
 - Spreadsheet tools for comparison and visualization
- Industry Technologies (Analyzed in Study):
 - IoT-based remote monitoring systems
 - Digital twins for asset performance
 - Cloud-based safety reporting tools

- Based on **High-Reliability Organization (HRO)** theory
- Shift from:
- Local disruptions → Global crisis resilience
- Increased importance of:
- IoT, AI, remote monitoring
- Emerging challenges:
- Digital overload
- Reduced human interaction
- Growing focus on:
- Psychological safety
- Workforce engagement

- Focus mainly on:
- Technical and compliance aspects
- Limited coverage of:
- Behavioral and human factors
- Workforce fragmentation
- Technology-focused studies ignore:
- Usability issues
- Mental health impact
- Lack of long-term analysis of safety systems

- Existing studies focus mainly on **short-term emergency responses**, not long-term HSMS evolution
- Lack of research linking **deferred maintenance to safety risks and system failures**
- Digital transformation analyzed mostly from a **technical perspective**, ignoring human usability
- Insufficient attention to:
 - Cognitive overload
 - Alarm fatigue
- Worker adaptation challenges
- Psychosocial factors (stress, fatigue, isolation) not well integrated into safety frameworks
- Limited research on workforce fragmentation and decentralized operations
- Need for a **holistic socio-technical safety system approach**

Impact of COVID-19 on Oil & Gas

- Operations affected by:
- Lockdowns and travel restrictions
- Supply chain disruptions
- Workforce reduced to **skeleton crews**
- Key impacts:
- Increased workload
- Higher chances of human error
- Maintenance backlog increased by **25–40%**
- Increase in minor incidents during restart phase

Key Challenges

- Workforce fragmentation due to travel restrictions and reduced manpower
- Reduced supervision and absence of middle management leading to weaker safety control
- Increased workload and **task saturation**, forcing employees to multitask
- Growth in deferred maintenance creating hidden operational risks
- Emergence of “**Risk Debt**”, where systems degrade over time without intervention
- Disruption of informal safety mechanisms:
 - Peer checks
 - On-site mentoring
- Reduced situational awareness due to screen-based monitoring
- Psychological stress, fatigue, and isolation among workers
- Difficulty in maintaining safety culture and team cohesion

Results and Findings

- HSMS proved to be a **dynamic system**, not a fixed structure
- Digital tools played a critical role in maintaining operations
- However, a “**Digital Paradox**” emerged:
 - More data availability
 - Reduced human intuition and situational awareness
- Organizations with:
 - Strong leadership
 - Flexible safety frameworks performed better
 - Increased individual accountability improved safety awareness
- Deferred maintenance resulted in:
 - Increased minor incidents during restart phase
- Key insight:
 - Effective safety requires **balance between technology and human factors**



Thank You